

Bari Prioritized Research Program

Executive Summary

The most important research move for Bari is not more ingredient-by-ingredient analysis in isolation. It is a category-first intelligence system that explains why two products with similar labels can behave very differently once raw materials, processing, matrix structure, authenticity, and real-world outcome evidence are taken seriously. That conclusion is supported by the way food law is written in Israel, by Codex commodity standards across dairy, honey, cheese, and canned fish, by the International Olive Council's grading framework for virgin olive oil, and by the growing food-matrix literature showing that structure and processing can materially change physiological effects beyond nutrient totals alone. ¹

For Bari, the strongest near-term sequence is this: **Category Excellence Intelligence first, Authenticity and Fraud Intelligence second, Food Matrix Intelligence third, and Outcome Intelligence fourth.** That is not because outcomes matter least. It is because product-level outcome inference is the hardest part to make defensible, while category standards, fraud controls, sensory frameworks, and raw-material/process markers are already operationally usable today. Put differently: Bari should first learn to identify what a *real, excellent, well-made* product is within a category before it tries to infer what that product likely means for long-term health. ²

The Israeli context makes this sequencing even more important. Israel's food law explicitly centers **quality, integrity, and safety**; the State Comptroller reported that by 2025 the reform had adopted 44 European regulations, including for sensitive foods such as olive oil, honey, and chocolate; and the same report noted gaps in mapping Israeli laboratory capability under the newer regime. Meanwhile, the Ministry of Health has continued to evolve front-of-pack and allergen labeling rules, and official Israeli sources show strong import dependence in several Bari-relevant categories. ³

If Bari wants the most useful pilot set, the best initial trio is **olive oil, yogurt, and bread.** Olive oil is the clearest combined test of category excellence, fraud detection, and outcome relevance. Yogurt is the clearest food-matrix case, with strong standards for identity and live cultures plus meaningful human evidence. Bread is strategically important because it is ubiquitous, highly processed in ways that matter, nutritionally over-simplified by most consumer systems, and highly sensitive to grain integrity, fermentation, and sodium. Honey should be a **phase-two pilot**, not phase one, because authenticity methods remain analytically and regulatorily contested, especially around NMR and reference-database adequacy. ⁴

Strategic Thesis

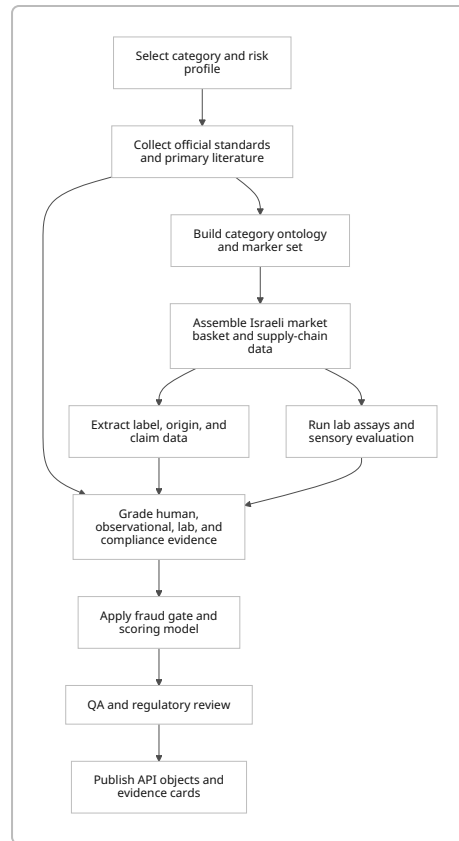
Bari should treat these four streams as one connected research system, not four separate projects. Category excellence without authenticity misses fraud. Authenticity without matrix analysis misses meaningful differences between otherwise compliant products. Matrix analysis without outcome intelligence risks becoming mechanistic but not actionable. Outcome intelligence without category and

integrity controls becomes too abstract and too easy to overclaim. The structural insight is that **Bari's edge will come from linking legally-defined food identity, manufacturing reality, lab evidence, and conservative outcome inference into one product knowledge graph.** ⁵

This also challenges a weak but common assumption: that the “best” research program is the one with the most biomedical endpoints. For Bari, that would be premature. The strongest early defensibility comes from proving that a product is genuinely what it claims to be, is made from strong raw materials, follows process patterns associated with better category quality, and maps to higher-confidence matrix and outcome evidence only where such inference is justified. WHO guidance, food-matrix reviews, and GRADE-style certainty frameworks all point in the same direction: nutrition decisions improve when foods are evaluated as foods, not as disconnected nutrient atoms or marketing claims. ⁶

The recommended Bari priority stack is shown below.

Research stream	Defensibility impact	Data maturity today	SKU-level tractability	Bari priority
Category Excellence Intelligence	Very high	High	High	First
Authenticity and Fraud Intelligence	Very high	Medium-high	High in priority categories	Second
Food Matrix Intelligence	High	Medium	Medium	Third
Outcome Intelligence	High	Medium-high at food-group level, lower at SKU level	Low-medium	Fourth



The Israeli market is especially suitable for this architecture because high-relevance categories combine strong local demand with import dependence, evolving EU-aligned rules, and categories where product integrity matters materially. Official Israeli data reported that imports supplied 97% of grains and grain products and 91% of fish in the 2020 food balance sheet, while the Ministry of Agriculture said Turkish imports accounted for about 83% of Israel's olive-oil imports in 2024. That means Bari should treat supply-chain intelligence as core infrastructure, not as a later add-on. 7

Research Streams, Questions, and Methods

Stream	Definition and scope	Key research questions and hypotheses	Methodologies	Primary and official data sources
Category Excellence Intelligence	Bari-defined category-specific intelligence on raw-material quality, legal identity, manufacturing discipline, analytical compliance, and sensory excellence. It asks what “best in class” actually means <i>within</i> bread, yogurt, olive oil, honey, tuna, and other categories, rather than across food categories that are not directly comparable. The conceptual foundation comes from commodity standards and category-specific sensory methods, not from generic nutrition panels alone. ⁸	Questions: What markers best distinguish excellent from merely compliant products? Which raw-material and process features actually predict better sensory and analytical performance? Hypothesis: in most categories, within-category quality variance is larger and more informative than generic “healthy/not healthy” labeling alone. ⁹	Build category ontologies; extract legal identity and claims; run risk-based market baskets; compare products against Codex/IOC/SCA/ICCO/AACC methods; collect blinded expert sensory data; create marker libraries and threshold rules.	Codex standards; IOC methods; SCA Coffee Value Assessment; AACC methods and definitions; Cacao of Excellence and ICCO guidance; Israeli Ministry of Health recall and labeling pages; State Comptroller import reports; GS1 Israel product data infrastructure. ¹⁰

Stream	Definition and scope	Key research questions and hypotheses	Methodologies	Primary and official data sources
Food Matrix Intelligence	Bari-defined intelligence on the structural and physicochemical properties of foods that influence digestion, bioavailability, satiety, glycemic response, and metabolic effects beyond nutrient totals. The food-matrix literature, especially in dairy, fermented foods, and grain structure, is the main scientific base. ¹¹	Questions: Which matrix attributes can be operationalized at product level? When does fermentation, intact structure, or concentrated protein meaningfully change likely effects? Hypothesis: fermentation, intact grain structure, and higher structural integrity explain more of a product's likely physiological behavior than single nutrients in isolation , especially in yogurt, cheese, and bread. ¹²	Evidence mapping of human trials and mechanistic studies; in-vitro digestion proxies; viscosity/syneresis/ textural testing; starch digestibility and estimated glycemic response work; small cross-over CGM and satiety pilots; category-specific matrix archetypes.	PubMed and PMC primary papers; WHO healthy-diet guidance; FoodData Central for nutrient baselines; PubChem for compound lookup; Codex fermented-milk and cheese identity standards. ¹³

Stream	Definition and scope	Key research questions and hypotheses	Methodologies	Primary and official data sources
Authenticity and Fraud Intelligence	Bari-defined intelligence on economically motivated adulteration, mislabeling, origin fraud, undeclared substitution, dilution, and suspicious blends. FDA defines food fraud as economically motivated adulteration, and EU/JRC work shows that high-value categories require targeted authenticity methods rather than simple trust in front labels. ¹⁴	Questions: Which categories are most exposed? Which analytical methods are sufficiently validated for Bari to rely on? Hypothesis: a relatively small set of categories—especially olive oil, honey, tuna/ canned fish, chocolate, and selected cheeses—drive a disproportionate share of trust risk, and price/origin/claim anomalies can improve sampling efficiency. ¹⁵	Vulnerability assessment by category; risk-based SKU and lot sampling; chain-of-custody controls; isotopic, spectroscopic, chromatographic, and DNA methods where fit-for-purpose; monitoring of recalls and official actions; “suspected vs confirmed” separation.	FDA EMA pages; JRC food-authenticity materials; EU “From the Hives” action; FSA honey authenticity work; RASFF; Israeli Ministry of Health recall portal and category-specific notices; ISRAC-accredited labs. ¹⁶

Stream	Definition and scope	Key research questions and hypotheses	Methodologies	Primary and official data sources
Outcome Intelligence	Bari-defined evidence system translating dietary-pattern, food-group, and product-relevant studies into conservative “outcome fit” signals. This stream should estimate <i>probable fit with better dietary patterns and lower-risk exposure patterns</i> , not make disease-treatment claims. ¹⁷	Questions: What outcomes are defensible at SKU level? How should Bari separate direct evidence from extrapolation? Hypothesis: the safest outcome layer is not product-as-medicine, but product-as-better-or-worse fit within evidence-backed patterns , such as Mediterranean-pattern fats, fermented dairy, higher whole-grain integrity, and lower ultra-processed exposure. ¹⁸	Hierarchical evidence synthesis; human-intervention and cohort mapping; GRADE-based certainty scoring; category-specific causal models; explicit penalties for indirectness and over-extrapolation; limited user-facing language for uncertain signals.	PREDIMED and related trials; fermented-food intervention studies; whole-grain and satiety studies; UPF umbrella reviews and cohorts; WHO dietary guidance; GRADE, RoB 2, ROBINS-I, QUADAS-2 where relevant. ¹⁹

These streams should be fed by a shared Bari source stack. The most practical core is: PubMed E-utilities for biomedical literature, Crossref REST for citation metadata and links, Open Food Facts for secondary label coverage, USDA FoodData Central for nutrient and reference food data, PubChem PUG-REST for compounds and analytes, RASFF for official EU food-risk notifications, Israeli government recall and labeling pages, and GS1 Israel for structured product identifiers where access is feasible. Open Food Facts should be used as a coverage layer, not as source of truth when official packaging, lab data, or government records are available. ²⁰

Category Intelligence Tables

The table set below is the operational heart of the program. These are not consumer-facing claims. They are **research markers** Bari can use to build category dossiers, sampling plans, assays, and scoring logic. The purpose is to replace vague “healthy ingredients” heuristics with category-native definitions of excellence, integrity, and likely matrix behavior. Codex, IOC, SCA, ICCO, and AACC frameworks are especially useful because they define what the product *is*, how it is *measured*, and in some cases how it is *sensory-graded*. ²¹

Category	Raw-material quality markers	Manufacturing and process markers	Product-integrity and authenticity markers	Measurable lab and analytical tests	Consumer sensory and quality indicators	Suggested data sources
Bread	Grain type; true whole-grain retention; flour ash/protein; kernel integrity; starter quality	Fermentation length; sourdough vs straight dough; pH and total titratable acidity; milling fineness; sodium	“Whole grain” claim verification; “sourdough” claim verification; refined flour substitution behind artisanal language	Protein; ash; fiber; sodium; pH/TTA; moisture; texture; in-vitro starch digestibility; phytate if relevant	Crust aroma; crumb openness; chew; moisture balance; staling curve; perceived satiety	AACC whole-grain definition; sourdough quality and glycemic-response studies. 22
Butter	Cream quality; pure milkfat; feed and season effects where available	Cultured vs sweet cream; churning and washing; salt addition; storage and oxidation control	Minimum milkfat and moisture compliance; non-dairy fat substitution; salt declaration accuracy	Milkfat %; moisture %; salt %; acid value; peroxide value; fatty acid profile; volatile profile by GC-MS	Spreadability at refrigeration; clean lactic/diacetyl notes; absence of rancidity; mouth-coating quality	Codex butter standard; cultured vs sweet cream butter flavoromics and volatile studies. 23
Yogurt	Milk solids; protein level; milk source; added sugar level; starter-culture identity	Fermentation endpoint; live-culture viability; straining/concentration; heat treatment after fermentation; stabilizer load	Correct yogurt identity; live-culture or probiotic claim verification; distinction from heat-treated fermented dessert	CFU count through shelf life; pH; titratable acidity; protein; sugar; viscosity; syneresis; texture profile	Clean tartness; body and thickness; low wheying-off; smoothness; balanced sweetness-acidity	Codex fermented milks; IDFA Live & Active Cultures material; yogurt systematic review and clinical studies. 24

Category	Raw-material quality markers	Manufacturing and process markers	Product-integrity and authenticity markers	Measurable lab and analytical tests	Consumer sensory and quality indicators	Suggested data sources
Cheese	Milk type and quality; fat/protein balance; starter and adjunct cultures; species identity	Curd handling; brining; ripening time and environment; rind and microbial surface management	Variety/ PDO-style compliance where claimed; analogue-cheese substitution; species/ origin mismatch; shredded-cheese additive load	Moisture on fat-free basis; fat in dry matter; salt; proteolysis/ lipolysis markers; VOC profile; texture; microbiology	Aroma complexity; firmness/ elasticity or crumbliness; rind quality; meltability where relevant	Codex cheese standard; cheese VOC/ ripening literature; dairy-matrix trials comparing cheese and butter. ²⁵
Category	Raw-material quality markers	Manufacturing and process markers	Product-integrity and authenticity markers	Measurable lab and analytical tests	Consumer sensory and quality indicators	Suggested data sources
Olive oil	Cultivar; harvest timing; healthy fruit; low fruit damage; rapid milling after harvest	Extraction temperature and delay; oxygen/light control; filtration and storage discipline; bottle protection	EVOO grade compliance; origin/blend accuracy; seed-oil dilution; deodorized oil; false “cold pressed/extra virgin” claims	Free acidity; peroxide; UV absorbance; IOC sensory panel; DAGs; PPP; fatty-acid profile; sterols; waxes; phenolics	Fruitiness; bitterness; pungency; freshness; absence of rancid, fusty, musty defects	IOC sensory and chemistry methods; JRC olive-oil authenticity review; UC Davis freshness work; Israeli official recall signals. ²⁶

Category	Raw-material quality markers	Manufacturing and process markers	Product-integrity and authenticity markers	Measurable lab and analytical tests	Consumer sensory and quality indicators	Suggested data sources
Hummus	Chickpea cultivar and texture; tahini quality; acid source; fat source; salt	Soaking/ cooking adequacy; dehulling; tahini ratio; emulsification; acidification; cold-chain stability; preservative strategy	Chickpea:tahini integrity; lower-value oil substitution; shelf-life and preservative burden; RTE microbiological integrity	pH; water activity; total plate count; yeast/mold; pathogen panel; fat profile; texture; proximate composition; heavy metals in tahini where relevant	Smoothness; creaminess; graininess; bitterness from tahini; freshness; garlic-lemon balance	Chickpea/hummus nutrition literature; hummus shelf-life studies; commercial hummus quality work; tahini composition/ safety studies. ²⁷
Tuna and canned fish	Species identity; raw fish quality fit for fresh sale; packing-medium quality	Time-temperature control before canning; retort adequacy; fill and drained weight control; oil/ brine choice	Species substitution; origin mislabeling; weight inflation; sustainability/ traceability claim verification	Histamine; mercury; DNA mini-barcoding; salt; drained weight; oxidation markers; commercial sterility checks	Firm flakes; clean aroma; low mushiness; balanced salt; absence of metallic or stale notes	Codex canned tuna standard; FDA histamine guidance; FDA/EPA or EU mercury material; canned-tuna DNA barcoding studies. ²⁸
Honey	Mature honey; floral source; moisture; pollen profile; beekeeper and geography metadata	Heating intensity; filtration degree; blending; storage; crystallization control	Exogenous sugar addition; false floral/ origin claims; over-filtration removing pollen; adulterated blends	Moisture; HMF; diastase; conductivity; pollen analysis; EA/ LC-IRMS; HRMS; NMR only with reference-database caution	Floral specificity; crystallization pattern; aroma persistence; absence of burnt or caramelized notes	Codex honey standard; EU “From the Hives”; FSA on NMR limitations; IRMS method papers. ²⁹

Category	Raw-material quality markers	Manufacturing and process markers	Product-integrity and authenticity markers	Measurable lab and analytical tests	Consumer sensory and quality indicators	Suggested data sources
Chocolate	Cocoa mass percentage; bean origin; fermentation quality; cocoa-butter content; sugar quality	Fermentation; drying; roasting; conching; tempering; alkalization where used	Cocoa-solids accuracy; non-cocoa fat use; origin or fine-flavour claim inflation; heavy-metal contamination	Cadmium and lead; fat profile; particle size; rheology; polyphenols; volatile fingerprint; bloom stability	Snap; gloss; melt; bitterness-acidity balance; aftertaste; absence of stale or smoky defects	ICCO fine-flavour materials; Cacao of Excellence guide; EU cadmium pages; cocoa fermentation science. ³⁰
Pasta	Durum semolina purity; protein; yellow pigment; vitreousness; granulometry	Extrusion conditions; drying temperature/time; semolina hydration; added-fiber/protein formulation effects	"100% durum" and whole-grain claim integrity; fortification accuracy; common-wheat substitution in premium claims	Protein; ash; color; cooking loss; firmness; water absorption; texture; microstructure	Al dente firmness; low stickiness; color retention; sauce adhesion; breakage resistance	AACC semolina and pasta methods; semolina quality reviews; pasta-processing studies. ³¹
Cereal	Whole-grain quantity and type; beta-glucan sources; nut/seed inclusion; sugar source	Extrusion/flaking severity; toasting; particle size; starch gelatinization	Whole-grain claim integrity; added-sugar/sodium positioning; fortification label accuracy	Fiber; added sugar; sodium; starch digestibility; beta-glucan; texture and crispness retention	Crispness in milk; grain aroma; sweetness balance; portion satiety	FDA updated healthy rule for cereals; AACC whole-grain definition; cereal glycemic-response studies. ³²

Category	Raw-material quality markers	Manufacturing and process markers	Product-integrity and authenticity markers	Measurable lab and analytical tests	Consumer sensory and quality indicators	Suggested data sources
Coffee	Bean species and origin; moisture within quality range; defect count; bean size and uniformity	Roast degree and profile; rest period; grind consistency; packaging and degassing management	Origin/grade claim integrity; defect grading accuracy; stale dated stock; blend misdescription	Moisture; defect count; screen size; acidity/ chlorogenic acids; volatile profile; SCA CVA sensory assessment	Fragrance/ aroma; sweetness; clarity; acidity; body; uniformity; clean finish	ICO quality standard; FAO/ICO quality interpretation; SCA CVA standards; coffee-acid and roast science. ³³

Evidence-Grade Framework and Bari Scoring

Bari should not use one undifferentiated “evidence score.” It needs an **evidence-grade agent** that knows what kind of claim is being asked. Human outcome claims require one standard of proof. Authenticity claims require another. Test-method validity requires another. The clean approach is to combine GRADE for certainty of evidence, RoB 2 for randomized trials, ROBINS-I for non-randomized studies, QUADAS-2 for diagnostic-test accuracy, and ISO/IEC 17025 plus official-control frameworks for laboratory competence.

³⁴

Evidence class	Typical sources	Quality framework	Bari use case	Recommended ceiling
Human intervention evidence	RCTs, cross-over feeding studies, shelf-life-exposure pilots	GRADE + RoB 2	Outcome intelligence; selected food-matrix questions	Highest for direct outcome fit when category and dose match Bari use case
High-quality observational evidence	Prospective cohorts, repeated-measures population studies	GRADE + ROBINS-I logic for bias and confounding	Outcome fit where RCTs are impractical; long-term pattern evidence	Moderate; downgraded for indirectness at product level

Evidence class	Typical sources	Quality framework	Bari use case	Recommended ceiling
Validated laboratory and analytical evidence	ISO 17025 labs; Codex/IOC/AOAC/JRC-aligned methods; authenticated reference sets	QUADAS-2 where method accuracy is relevant; ISO 17025 competence	Authenticity, category excellence, shelf-life and safety markers	Highest for integrity and analytical truth, but not for long-term health claims
Official compliance and regulatory evidence	Recalls, border actions, official controls, standards compliance documents	Official-control and legal conformity review	Hard safety and fraud gates; legal identity	Can trigger hard caps or red flags regardless of other scores
Expert sensory evidence	IOC panels, SCA CVA, trained cheese/chocolate panels, blinded category tasting	Protocol adherence + panel reliability checks	Category excellence and consumer-quality prediction	Strong for excellence; weak for outcome claims
Certification and label evidence	PDO/PGI, organic, MSC, LAC seal, other third-party marks	Document audit only	Traceability prior, not proof by itself	Never above moderate without supporting data
Mechanistic evidence	In-vitro digestion, VOC studies, compositional chemistry, microbiome mechanistic work	Internal Bari mechanistic rubric	Food matrix and explanatory support	Supportive only unless concordant with human or validated analytical evidence

The core Bari scoring architecture should therefore be **multi-dimensional and gated**, not averaged into one false-precision number.

Proposed scoring dimensions

Dimension	What it measures	Default weight	Notes
Category Excellence Score	Raw-material quality, process discipline, sensory and category-native analytical quality	35%	Bari's main differentiation layer
Food Matrix Score	Structural and fermentation features likely to influence nutrient delivery, satiety, glycemic response, or microbial activity	25%	Most important in dairy, bread, cereal, pasta, hummus

Dimension	What it measures	Default weight	Notes
Integrity Score	Authenticity confidence, legal identity, traceability, fraud risk, and compliance posture	25%	Should rise to 35% in olive oil, honey, tuna, chocolate
Outcome Fit Score	Conservative match to favorable dietary-pattern evidence and absence of strong harm signals	15%	Kept lighter because SKU-level causality is harder

Critical rule: if Bari has credible evidence of adulteration, dangerous contamination, or major legal-identity failure, the product should be **hard-gated** into a capped or red state rather than “rescued” by favorable macros or marketing. That principle is consistent with FDA’s framing of food fraud and with the way official food law is built around product integrity and safety. ³⁵

The evidence-grade agent should output a structured object for every study or analytical source:

- claim type
- category match
- exposure specificity
- population match
- duration and dose realism
- risk of bias
- method validation status
- accredited-lab status
- consistency with other evidence
- directness for product scoring
- conflict-of-interest note
- Bari grade: **A, B, C, D**, plus a confidence interval or confidence band

A practical interpretation layer would work like this:

Bari grade	Meaning	Typical examples
A	Strong, direct, decision-ready evidence	Multiple consistent RCTs or validated official analytical methods with competent labs
B	Useful but partly indirect evidence	Good cohort evidence, single well-run cross-over trial, solid non-official but replicated method
C	Signal-generating evidence	Mechanistic work, early lab method, small human pilot, strong certification without direct proof
D	Too weak or too contested for scoring leverage	Unvalidated authenticity methods, poorly matched biomarker studies, uncontrolled marketing claims

For Bari scoring, certifications should be treated as **priors**, not endpoints. A live-cultures seal, PDO, or sustainability mark may improve the probability that a product is higher quality or better traced, but it should not replace direct analytical confirmation where authenticity or composition is central. That especially matters in honey, olive oil, and tuna. ³⁶

Deliverables, Timelines, Ownership, and Prompts

The program should be designed to produce reusable assets, not just one report. The right outputs are structured datasets, lab protocols, evidence agents, and score objects that can run continuously as the Israeli and European regulatory environment changes. That is especially important now that Israel has expanded EU-aligned food regulation while still dealing with laboratory-capability mapping and import-system reform. ³⁷

Priority	Deliverable	What Bari gets	Owner	Why this owner	Timeline	Effort
Critical	Category ontology and marker dictionary	One normalized schema for all 12 categories, including raw-material fields, process markers, integrity flags, assays, and sensory descriptors	Head of Product	This is a product architecture and prioritization decision before it is a research task	Weeks 1–3	Medium
Critical	Source registry and ingestion stack	Managed source map for PubMed, Crossref, Open Food Facts, FoodData Central, PubChem, RASFF, Israeli MoH/gov.il, GS1 Israel	Frontend Architect	It is primarily a data-engineering and systems-integration problem	Weeks 1–4	High

Priority	Deliverable	What Bari gets	Owner	Why this owner	Timeline	Effort
Critical	Bari Evidence Agent	Automated pipeline to classify study type, relevance, certainty, method validity, and conflicts	Research Analyst with Chief Nutrition Officer	Research Analyst should gather and normalize the evidence; CNO should define nutrition and outcome logic	Weeks 2–6	High
High	Fraud vulnerability engine	Category risk model, sampling logic, “suspected vs confirmed” states, and lot-tracking rules	QA & Audit Lead	Fraud outputs need audit discipline, chain-of-custody logic, and conservative escalation	Weeks 3–8	High
High	Assay library and lab partner roster	Approved assay menu, preferred methods, reference ranges, and ISRAC/ISO 17025 partner shortlist	Chief Nutrition Officer with QA & Audit Lead	Method relevance and lab reliability must be jointly owned	Weeks 3–7	High
High	Twelve category dossiers	Reusable research packets for the 12 example categories, each with benchmark tiers, known fraud vectors, and test menus	Research Analyst	This is primary research and synthesis work	Weeks 4–10	Medium

Priority	Deliverable	What Bari gets	Owner	Why this owner	Timeline	Effort
High	Scoring engine v1	Weighted dimension scores with integrity gate, category overrides, and explanation rules	Frontend Architect with Head of Product	Requires product logic plus implementation	Weeks 6–10	High
Medium	Pilot dashboard	A live internal console showing score dispersion, assay results, recalls, and evidence confidence	Frontend Architect	Engineering-led operational output	Weeks 8–12	Medium
Medium	Consumer evidence card design	User-facing explanation object showing why Bari scored a product the way it did	Design Director	Needed after score object and evidence object are stable	Weeks 10–14	Medium
High	Regression and audit suite	Test harness for citation integrity, scoring consistency, hallucination control, and regulatory wording	QA & Audit Lead	This is a validation problem first and foremost	Weeks 8–14	Medium

The external and semi-open data stack should be explicit from day one: PubMed E-utilities, Crossref REST, FoodData Central API, PubChem PUG-REST, Open Food Facts API, RASFF Window, Israeli recall and labeling pages, and GS1 Israel data infrastructure where commercial access is warranted. Bari should also maintain a curated folder of primary standards such as Codex, IOC, SCA, ICCO, and relevant Israeli government pages, because primary standards are often more important than general literature in category-quality work. ³⁸

Ownership prompts

Assessment	Owner	Why this owner	Copy-paste prompt
Freeze the research architecture before adding more categories	Head of Product	Priority, sequencing, and scope control sit here	Prompt: "Create Bari v1 research architecture for Category Excellence, Authenticity/Fraud, Food Matrix, and Outcome Intelligence. Deliver: final stream definitions, score-dimension boundaries, category-priority list, and phase-gate criteria for moving a category from research to production. Keep outputs implementation-ready and resolve any overlap between streams."
Build the evidence corpus and category dossiers	Research Analyst	This is the primary synthesis backbone	Prompt: "Build 12 Bari category dossiers for bread, butter, yogurt, olive oil, hummus, cheese, chocolate, tuna/canned fish, pasta, cereal, coffee, and honey. For each, extract category identity rules, raw-material markers, process markers, integrity risks, validated assays, sensory indicators, and strongest primary sources. Output in structured JSON plus a short narrative memo per category."
Define what counts as matrix and outcome evidence	Chief Nutrition Officer	This requires scientific judgment on food structure and physiological relevance	Prompt: "Draft Bari's v1 food-matrix and outcome-evidence rubric. Define which product features count as matrix-relevant, which outcomes are direct versus indirect, how to penalize over-extrapolation, and how to grade studies using GRADE-style certainty plus category relevance. Output explicit scoring rules, not prose only."
Implement the ingestion and scoring objects	Frontend Architect	This is engineering-intensive and depends on stable schemas	Prompt: "Implement Bari research data model v1. Required objects: Product, CategoryMarker, AssayResult, EvidenceItem, IntegrityFlag, ScoreDimension, and SourceRecord. Add connectors for PubMed, Crossref, FoodData Central, PubChem, Open Food Facts, RASFF, and Israeli official recall/label pages. Return typed schemas, endpoint plan, and sync cadence."

Assessment	Owner	Why this owner	Copy-paste prompt
Prevent false confidence and claim drift	QA & Audit Lead	Bari's credibility will fail if fraud flags and evidence grades are not audited	Prompt: "Design Bari's research QA suite. Include duplication checks, citation validation, method-validation checks, suspected-versus-confirmed fraud logic, regulatory wording controls, and regression tests for scoring stability across the 12 category dossiers. Output pass/fail criteria and weekly audit checklist."
UI work should wait until the evidence object is stable	Design Director	Design is important, but too early here creates expensive rework	No prompt needed yet.

Risks, Limitations, and Regulatory Considerations

The largest scientific risk is **over-translation**. Bari can responsibly score category excellence and integrity earlier than it can claim long-term health effects. Food-matrix science is real and increasingly important, but the strongest evidence remains uneven by category, and long-term outcomes often come from dietary-pattern studies rather than SKU-level trials. That is why Bari should expose outcome intelligence as a confidence-weighted fit signal, not as a disease claim engine. ³⁹

The largest analytical risk is **method overconfidence in authenticity testing**. Olive oil and honey are especially instructive. JRC has highlighted persistent olive-oil authenticity and quality issues, but also the importance of appropriate analytical approaches. In honey, the EU coordinated action showed that imported consignments and operators could be flagged as suspicious, yet the same EU and FSA materials make clear that validated official methods remain a live issue and that NMR in particular is widely used but disputed as a yes/no regulatory tool. Bari therefore needs a method-status layer: validated official, validated but limited, promising non-official, and contested. ⁴⁰

The largest operational risk in Israel is **regulatory transition plus laboratory capacity**. The State Comptroller reported that the 2025 reform had incorporated 44 European regulations, including sensitive foods such as olive oil, honey, and chocolate, while also noting that laboratory-testing capability for the newer framework had not been comprehensively mapped. That means Bari should assume that official compliance infrastructure is improving but still uneven, and should preferentially work with ISRAC-recognized or ISO/IEC 17025-accredited labs and keep strict chain-of-custody records. ⁴¹

The most important Israeli regulatory considerations are straightforward and highly relevant to product intelligence. Israel's Public Health Protection (Food) Law is explicitly built around quality, integrity, safety, and traceable manufacturer/importer responsibility. The Ministry of Health has continued to refine food labeling, including allergen labeling changes from January 2025 and continued development of the dual red/green front-of-pack system. Those rules create a powerful structured signal for Bari, but they do not replace category-native quality or authenticity work. A bread can avoid a red label and still be poor bread; an olive oil can look premium and still fail integrity. ⁴²

The central ethical risk is **unfair brand harm**. Bari should never conflate “evidence missing” with “evidence negative,” and it should never present a fraud signal as established fact unless the underlying lab method is fit for purpose, the sample chain is secure, and the finding is internally reviewed. This is especially important for artisanal Israeli producers, smaller importers, and categories with analytically disputed methods such as honey. At the same time, Bari should not give large manufacturers a free pass simply because they are more heavily documented. Equal skepticism is part of the design. ⁴³

Pilot Study Designs

The proposed pilot set is **olive oil, yogurt, and bread**. That trio is the best balance of strategic importance, matrix relevance, analytical tractability, fraud exposure, and likely consumer value. Honey is important, especially in Israel where there are about 120,000 registered hives and strong local-production relevance, but it should follow after Bari has built stronger method-governance around reference databases and disputed authenticity approaches. ⁴⁴

Pilot category	Why it should be first-wave	Proposed sample	Proposed assays and measurements	Primary metrics	Timeline	Expected Bari decision
Extra virgin olive oil	Best combined test of category excellence, fraud intelligence, and dietary-pattern relevance. Also highly relevant in Israel because of recent official scrutiny and import dependence. ⁴⁵	60 SKUs, two lots each, split across Israeli producers, EU imports, and non-EU imports; 8–12 trained tasters following IOC-compatible panel rules	Free acidity, peroxide, K232/K270/ΔK, IOC sensory panel, polyphenols, DAGs, PPP, fatty-acid profile, sterols, waxes, packaging audit, label/origin review	EVOO pass rate; integrity-fail rate; freshness dispersion; Israeli vs imported score spread; chemistry-sensory agreement	14 weeks	Finalize Bari’s v1 integrity gate, freshness subscore, and olive-oil explanation card

Pilot category	Why it should be first-wave	Proposed sample	Proposed assays and measurements	Primary metrics	Timeline	Expected Bari decision
Yogurt	Strongest matrix pilot. Identity is well-defined, live-culture relevance is testable, and there is enough human evidence to support conservative outcome-fit logic. ⁴⁶	40 SKUs , two lots each, across plain, strained, flavored, and drinkable products; nested 30-adult cross-over pilot for acute satiety and CGM outcomes	CFU at purchase and near end of shelf life, pH, titratable acidity, protein, sugars, viscosity, syneresis, texture; short cross-over measurements of postprandial glucose and satiety VAS for 3–4 product archetypes	Culture survival; protein-to-sugar quality tradeoff; satiety score; CGM iAUC; sensory acceptance; proportion of products with “live culture” evidence matching claims	16 weeks	Lock Bari’s matrix rules for fermented dairy and define how live-culture, protein, and sugar interact in score logic
Bread	Bread is too important to leave to generic label logic. It is also where Bari can show the value of grain integrity and fermentation rather than simplistic carb fear. ⁴⁷	45 SKUs , two lots each, spanning refined, whole-grain, and sourdough-positioned products; nested 24-adult cross-over pilot comparing 3–4 bread archetypes	Sodium, fiber, moisture, pH/TTA, texture, crumb imaging, in-vitro starch digestibility, ingredient audit for whole-grain and sourdough plausibility; short satiety and CGM pilot	Estimated starch-hydrolysis ranking; satiety VAS; sodium-normalized quality score; authenticity gap between positioning and measured characteristics; staling curve	16 weeks	Build Bari’s bread archetypes and prove that matrix-aware scoring beats conventional nutrient-panel ranking

Several design choices matter here.

First, Bari should sample **lots**, not just SKUs. Fraud, oxidation, and shelf-life effects are often batch-sensitive. Second, all analytical work should use **blinded coding**, and any score that affects public-facing trust should be reviewed by QA before release. Third, the human sub-studies in yogurt and bread should be explicitly framed as **directional matrix pilots**, not definitive health-effect trials. They are intended to calibrate Bari’s

matrix logic, not to create regulated health claims. That is consistent with the certainty principles built into GRADE and risk-of-bias frameworks. ⁴⁸

The right success criteria for wave one are not “publish a perfect score.” They are more concrete:

- Can Bari identify high-confidence fraud or non-identity risks in priority categories without overcalling them?
- Can Bari show that certain process and matrix markers explain meaningful quality differences inside categories?
- Can Bari move from label reading to evidence-backed category judgment with auditable provenance?
- Can Bari explain the result to a consumer or partner in a way that is scientifically cautious but commercially useful?

If those four questions are answered well, Bari will have built something much more valuable than another nutrition-summary layer: it will have built a defensible **food intelligence operating system**.

¹ ³ ⁵ ⁴² https://www.fao.org/fileadmin/user_upload/gmfp/docs/Israeli%20public%20health%20protection%202015.pdf

https://www.fao.org/fileadmin/user_upload/gmfp/docs/Israeli%20public%20health%20protection%202015.pdf

² ¹⁴ ¹⁶ ³⁵ <https://www.fda.gov/food/compliance-enforcement-food/economically-motivated-adulteration-food-fraud>

<https://www.fda.gov/food/compliance-enforcement-food/economically-motivated-adulteration-food-fraud>

⁴ ²⁶ ⁴⁵ <https://www.internationaloliveoil.org/wp-content/uploads/2019/11/COI-T20-Doc.-15-REV-10-2018-Eng.pdf>

<https://www.internationaloliveoil.org/wp-content/uploads/2019/11/COI-T20-Doc.-15-REV-10-2018-Eng.pdf>

⁶ <https://www.who.int/news-room/fact-sheets/detail/healthy-diet>

<https://www.who.int/news-room/fact-sheets/detail/healthy-diet>

⁷ מאזן אספקת המזון

https://www.cbs.gov.il/he/publications/DocLib/2022/1863/h_print.pdf?utm_source=chatgpt.com

⁸ ²⁵ [https://www.fao.org/fao-who-codexalimentarius/sh-proxy/hu?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/hu?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B283-1978%252FCXS%2B283-1978%252FCXS_283e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B283-1978%252FCXS_283e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/hu?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B283-1978%252FCXS_283e.pdf)

[https://www.fao.org/fao-who-codexalimentarius/sh-proxy/hu?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/hu?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B283-1978%252FCXS_283e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B283-1978%252FCXS_283e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/hu?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B283-1978%252FCXS_283e.pdf)

⁹ ³² <https://www.fda.gov/food/hfp-constituent-updates/fda-finalizes-updated-healthy-nutrient-content-claim>

<https://www.fda.gov/food/hfp-constituent-updates/fda-finalizes-updated-healthy-nutrient-content-claim>

¹⁰ ²¹ ²⁴ ⁴⁶ [https://www.fao.org/fao-who-codexalimentarius/sh-proxy/es/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/es/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B243-2003%252FCXS%2B243-2003%252FCXS_243e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B243-2003%252FCXS_243e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/es/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B243-2003%252FCXS_243e.pdf)

[https://www.fao.org/fao-who-codexalimentarius/sh-proxy/es/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/es/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B243-2003%252FCXS_243e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B243-2003%252FCXS_243e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/es/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B243-2003%252FCXS_243e.pdf)

¹¹ ³⁹ <https://pubmed.ncbi.nlm.nih.gov/30040431/>

<https://pubmed.ncbi.nlm.nih.gov/30040431/>

12 <https://pmc.ncbi.nlm.nih.gov/articles/PMC11397276/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC11397276/>

13 20 38 <https://www.ncbi.nlm.nih.gov/home/develop/api/>

<https://www.ncbi.nlm.nih.gov/home/develop/api/>

15 40 <https://publications.jrc.ec.europa.eu/repository/handle/JRC114784>

<https://publications.jrc.ec.europa.eu/repository/handle/JRC114784>

17 34 48 <https://www.gradeworkinggroup.org/>

<https://www.gradeworkinggroup.org/>

18 19 <https://www.nejm.org/doi/full/10.1056/NEJMoa1800389>

<https://www.nejm.org/doi/full/10.1056/NEJMoa1800389>

22 47 <https://www.cerealsgrains.org/resources/definitions/Documents/WholeGrains/wgflyer.pdf>

<https://www.cerealsgrains.org/resources/definitions/Documents/WholeGrains/wgflyer.pdf>

23 [https://www.fao.org/fao-who-codexalimentarius/sh-proxy/it/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/it/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B279-1971%252FCXS_279e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B279-1971%252FCXS_279e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/it/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B279-1971%252FCXS_279e.pdf)

[https://www.fao.org/fao-who-codexalimentarius/sh-proxy/it/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/it/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B279-1971%252FCXS_279e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B279-1971%252FCXS_279e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/it/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B279-1971%252FCXS_279e.pdf)

27 <https://pmc.ncbi.nlm.nih.gov/articles/PMC5188421/>

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5188421/>

28 [https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B70-1981%252FCXS_070e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B70-1981%252FCXS_070e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B70-1981%252FCXS_070e.pdf)

[https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B70-1981%252FCXS_070e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B70-1981%252FCXS_070e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B70-1981%252FCXS_070e.pdf)

29 [https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B12-1981%252FCXS_012e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B12-1981%252FCXS_012e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B12-1981%252FCXS_012e.pdf)

[https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B12-1981%252FCXS_012e.pdf)

[Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B12-1981%252FCXS_012e.pdf](https://www.fao.org/fao-who-codexalimentarius/sh-proxy/en/?Ink=1&url=https%253A%252F%252Fworkspace.fao.org%252Fsites%252Fcodex%252FStandards%252FCXS%2B12-1981%252FCXS_012e.pdf)

30 <https://www.icco.org/wp-content/uploads/FFP-5-2-Rev.1-A-Working-Definition-of-Fine-or-Flavour-Cocoa-English.pdf>

<https://www.icco.org/wp-content/uploads/FFP-5-2-Rev.1-A-Working-Definition-of-Fine-or-Flavour-Cocoa-English.pdf>

31 https://www.cerealsgrains.org/resources/Methods/Pages/66Semolina_Pasta_NoodleQuality.aspx

https://www.cerealsgrains.org/resources/Methods/Pages/66Semolina_Pasta_NoodleQuality.aspx

33 <https://www.ico.org/documents/iccres420e.pdf>

<https://www.ico.org/documents/iccres420e.pdf>

36 <https://www.idfa.org/live-active-cultures-seal>

<https://www.idfa.org/live-active-cultures-seal>

37 41 <https://library.mevaker.gov.il/sites/DigitalLibrary/Documents/2025/2025-10/EN/2025-102025.10-202-Import-Taktzir-EN.pdf>

<https://library.mevaker.gov.il/sites/DigitalLibrary/Documents/2025/2025-10/EN/2025-102025.10-202-Import-Taktzir-EN.pdf>

43 [Honey authenticity: the NMR issue | Food Standards Agency](https://www.food.gov.uk/research/honey-authenticity-the-nmr-issue)

<https://www.food.gov.uk/research/honey-authenticity-the-nmr-issue>

44 <https://www.gov.il/en/departments/topics/bees>
<https://www.gov.il/en/departments/topics/bees>